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(12) **United States Design Patent** (10) **Patent No.: US D1,093,551 S**
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(54) **COMPONENT MIXING CHAMBER**

B05B 1/02; B05B 1/26; B05B 12/002;
B05B 1/18; B05B 9/01

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See application file for complete search history.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

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MN (US)

675,813	A	6/1901	Carling	
3,289,502	A	12/1966	Jessen	
3,399,837	A	9/1968	Frick	
3,437,273	A	4/1969	Hagfors	
3,488,004	A	1/1970	Peeps	
3,708,123	A	1/1973	Krueger	
3,796,990	A	3/1974	Hill	
3,799,403	A	3/1974	Probst et al.	
3,837,575	A	9/1974	Lehnert	
3,901,408	A *	8/1975	Boden	G01F 11/029 222/134

(**) Term: **15 Years**

4,074,857	A	2/1978	Calder	
4,117,551	A	9/1978	Brooks et al.	
4,193,546	A	3/1980	Goelz et al.	
4,392,617	A	7/1983	Bakos et al.	
4,453,670	A	6/1984	Sirovy	
4,676,437	A	6/1987	Brown	
4,708,292	A	11/1987	Gammons	
4,715,537	A	12/1987	Calder	
4,913,317	A	4/1990	Wernicke	
5,004,351	A *	4/1991	Salaba	B29B 7/76 137/597 B29B 7/76 422/131
5,067,886	A *	11/1991	Salaba	

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(51) **LOC (15) Cl.** **23-01**

(52) **U.S. Cl.**

USPC **D23/213**

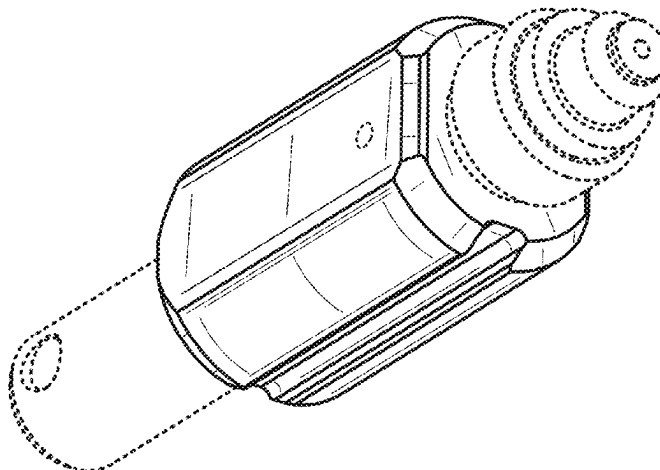
CPC **B05B 7/0408** (2013.01); **B01F 35/522**
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5,086,949	A	2/1992	Vulpitta et al.	
5,090,814	A	2/1992	Petcen	
5,104,006	A	4/1992	Brown	
5,129,581	A	7/1992	Braun et al.	
5,163,584	A	11/1992	Huber et al.	
5,219,097	A	6/1993	Huber et al.	
5,299,740	A	4/1994	Bert	
D356,853	S	3/1995	Fowler	
5,429,308	A	7/1995	Brown	
D366,051	S	1/1996	Lewis et al.	
5,609,302	A	3/1997	Smith	
5,829,680	A	11/1998	Perret	
5,884,847	A	3/1999	Christopher	
5,922,131	A	7/1999	Haas	
6,158,672	A	12/2000	Millan	
D436,648	S	1/2001	Maddox et al.	
6,264,113	B1	7/2001	Dingler	
6,283,329	B1	9/2001	Bezaire et al.	
6,312,293	B1	11/2001	Wang	
6,315,161	B1	11/2001	Bezaire et al.	



6,412,516	B1	7/2002	Goldsmith	
D470,818	S	2/2003	Cheng	
D471,563	S	3/2003	Selic et al.	
D481,102	S *	10/2003	Marshik	D23/213
6,685,106	B1	2/2004	Van	
6,820,315	B1	11/2004	Hans et al.	
6,824,075	B2	11/2004	Zimmermann	
6,884,230	B1	4/2005	Epstein et al.	
6,981,478	B2	1/2006	Schaefer et al.	
7,059,545	B2	6/2006	Reetz	
7,182,221	B2	2/2007	Hanna et al.	
7,222,753	B2	5/2007	Hayduk	
D558,673	S	1/2008	Easley	
D565,697	S	4/2008	Thiel et al.	
D567,900	S	4/2008	Patterson et al.	
7,377,404	B2	5/2008	Cherfane	
7,527,172	B2	5/2009	McMichael	
7,552,847	B2	6/2009	Hayduk	
7,661,606	B2	2/2010	Vacher	
7,694,893	B2	4/2010	Zittel et al.	
D617,870	S *	6/2010	Wright	B05B 3/002 D23/213
D636,844	S	4/2011	Herre et al.	
D639,905	S *	6/2011	Uchida	D23/213
D639,906	S	6/2011	Wright	
8,365,958	B2	2/2013	Ho et al.	
8,590,809	B2	11/2013	Escoto et al.	
8,684,035	B2	4/2014	Bernhard	
8,726,831	B2	5/2014	Mather et al.	
8,875,950	B2	11/2014	Hayduk	
8,899,501	B2	12/2014	Fox et al.	
8,931,715	B2	1/2015	Courier	
8,991,729	B2	3/2015	Scherer	
9,038,929	B1	5/2015	Commette	
D742,927	S	11/2015	Thompson et al.	
9,192,950	B2	11/2015	Carleton et al.	
D745,635	S	12/2015	Schwarz	
D752,648	S	3/2016	Kopel et al.	
9,433,955	B2	9/2016	Haralson	
D776,787	S *	1/2017	Zhang	D23/213
D812,714	S *	3/2018	Hughes	D23/213
D822,156	S *	7/2018	Krebs	D23/213
D872,828	S *	1/2020	Bloom	D23/213
D872,829	S *	1/2020	Bloom	D23/213
D875,883	S *	2/2020	Fujii	D23/213
D877,777	S	3/2020	Pickard	
10,610,879	B2	4/2020	Duong et al.	
D888,197	S	6/2020	McDonnell	
D888,890	S	6/2020	Hamlin	
10,702,877	B2	7/2020	Ellis et al.	
11,135,609	B2	10/2021	Corona et al.	
11,213,840	B2	1/2022	Owens et al.	
11,413,635	B2	8/2022	Ross et al.	
11,633,748	B2	4/2023	Ross et al.	
11,896,992	B2	2/2024	Ross et al.	
2003/0051870	A1	3/2003	Robertson	
2005/0035220	A1	2/2005	Brown	
2007/0034713	A1	2/2007	Wang et al.	
2007/0034716	A1	2/2007	Zittel et al.	
2007/0170282	A1	7/2007	McMichael	
2008/0257979	A1	10/2008	Crawford	
2008/0260889	A1	10/2008	Jenko et al.	
2010/0065130	A1	3/2010	Swab et al.	
2011/0062702	A1	3/2011	Pellin et al.	
2011/0100364	A1	5/2011	Faram	
2012/0181347	A1	7/2012	Scherer	
2013/0092247	A1	4/2013	Lee et al.	
2013/0140225	A1	6/2013	Decock et al.	
2013/0292491	A1	11/2013	Cvetezar et al.	
2013/0320110	A1	12/2013	Brose et al.	
2014/0048627	A1	2/2014	Schmon et al.	
2014/0103144	A1	4/2014	Haralson	
2014/0224898	A1	8/2014	Courier	
2015/0202637	A1	7/2015	Nikles	
2015/0361968	A1 *	12/2015	Schroeder	E01C 23/22 417/53
2016/0001307	A1	1/2016	Schurle	
2016/0008825	A1	1/2016	Drozd	
2016/0303595	A1	10/2016	Duncan et al.	

2016/0361734	A1	12/2016	Routen et al.
2018/0021611	A1	1/2018	Unrath
2018/0104854	A1	4/2018	Sinders et al.
2018/0126396	A1	5/2018	Ellis et al.
2020/0038889	A1	2/2020	Volk et al.
2020/0129998	A1 *	4/2020	Ross
2021/0154694	A1	5/2021	Masse

FOREIGN PATENT DOCUMENTS

CA	2713643	A1	2/2012
CA	2998798	A1	9/2018
CN	1700527	A	11/2005
CN	101537399	A	9/2009
CN	103769321	A	5/2014
EP	0063707	A1	11/1982
ES	2364029	T3	8/2011
GB	1381556	A	1/1975

OTHER PUBLICATIONS

Fifth Canadian Office Action for CA Application No. 3111295, Dated Nov. 2, 2023, pp. 3.
First Canadian Office Action for CA Application No. 3111295, Dated Apr. 28, 2021, pp. 4.
Third Canadian Office Action for CA Application No. 3111295, Dated Aug. 3, 2022, pp. 3.
First Chinese Office Action for CN Application No. 201980070293.1, Dated Feb. 9, 2022, pp. 19.
First Chinese Office Action for CN Application No. 201980070370.3, Dated Feb. 9, 2022, pp. 22.
First Examination Report for IN Application No. 326134-001, Dated Feb. 27, 2020, pp. 2.
First Examination Report for IN Application No. 326135-001, Dated Feb. 21, 2020, pp. 2.
First Examination Report for IN Application No. 326138-001, Dated Feb. 19, 2020, pp. 2.
First Examination Report for IN Application No. 326139-001, Dated Feb. 20, 2020, pp. 2.
Fourth Canadian Office Action for CA Application No. 3111295, Dated Jun. 13, 2023, pp. 3.
Graco, PC42FL Fusion ProConnect Flat 01 Mix Chamber with 0.042 in (1.1 mm) Orifice Size, (site visited Sep. 22, 2020), Graco.com, URL: <<https://www.graco.com/us/en/contractor/product/pc42fl-fusion-pc.html>> (Year: 2020).
<https://www.amazon.com/Graco-Fusion-Mixing-Chamber-AR5252/dp/B011P26D2U> (Year: 2015).
International Preliminary Report on Patentability for PCT Application No. PCT/US2019/058095, Dated Apr. 27, 2021, pp. 12.
International Preliminary Report on Patentability for PCT Application No. PCT/US2019/058101, Dated Apr. 27, 2021, pp. 10.
International Search Report and Written Opinion for PCT Application No. PCT/US2019/058095, Dated Feb. 7, 2020, pp. 17.
International Search Report and Written Opinion for PCT Application No. PCT/US2019/058101, Dated Feb. 7, 2020, pp. 14.
Second Canadian Office Action for CA Application No. 3111295, Dated Sep. 28, 2021, pp. 3.
Second Chinese Office Action for CN Application No. 201980070293.1, Dated Sep. 28, 2022, pp. 20.
Second Chinese Office Action for CN Application No. 201980070370.3, Dated Oct. 31, 2022, pp. 27.

* cited by examiner

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(57)

CLAIM

The ornamental design for a component mixing chamber as shown and described.

DESCRIPTION

FIG. 1 is an isometric view of a component mixing chamber;
FIG. 2 is a top plan view of the component mixing chamber;

FIG. 3 is a left side elevational view of the component mixing chamber;

FIG. 4 is a right side elevational view of the component mixing chamber;

FIG. 5 is a bottom plan view of the component mixing chamber;

FIG. 6 is a front elevational view of the component mixing chamber; and,

FIG. 7 is a rear elevational view of the component mixing chamber.

The broken lines in the figures show portions of a component mixing chamber and form no part of the claimed design.

1 Claim, 3 Drawing Sheets

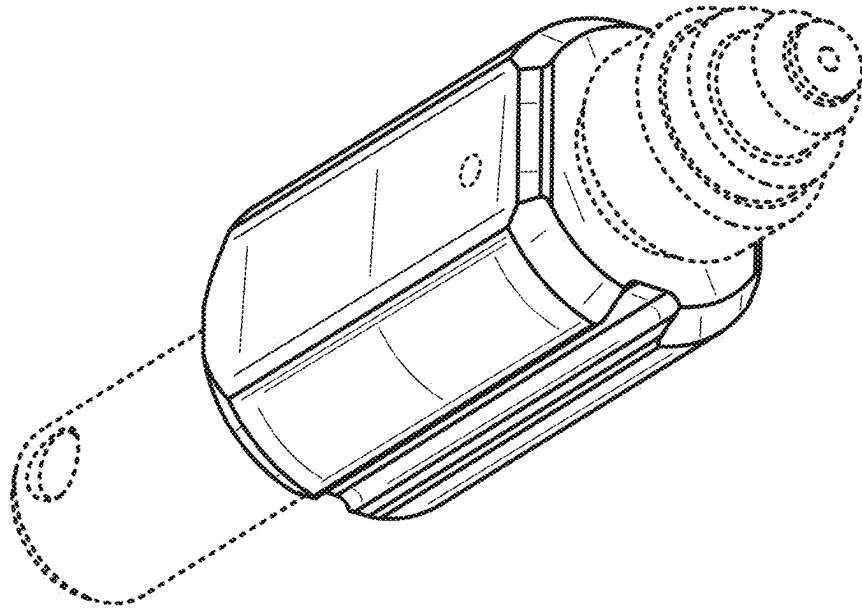


Fig. 1

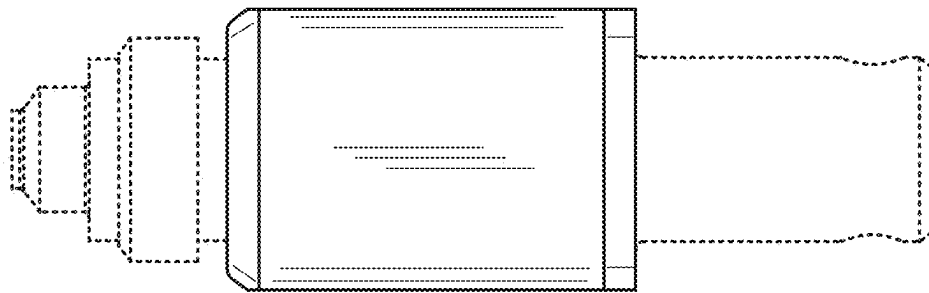


Fig. 2

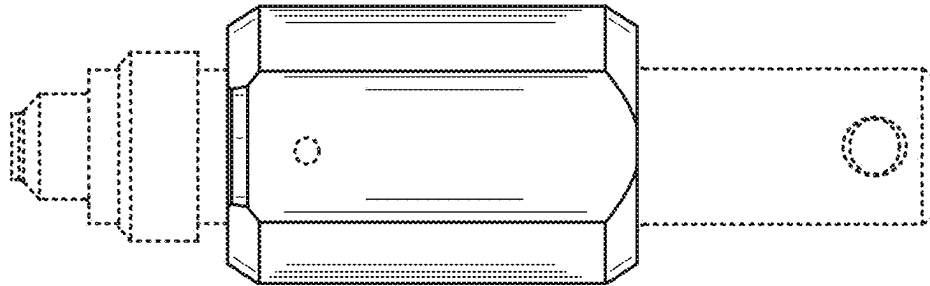


Fig. 3

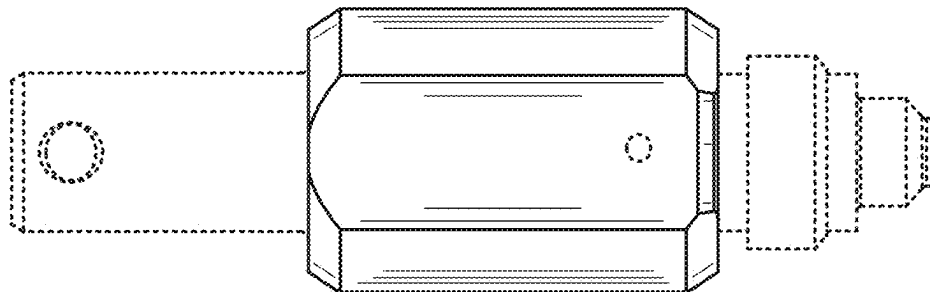


Fig. 4

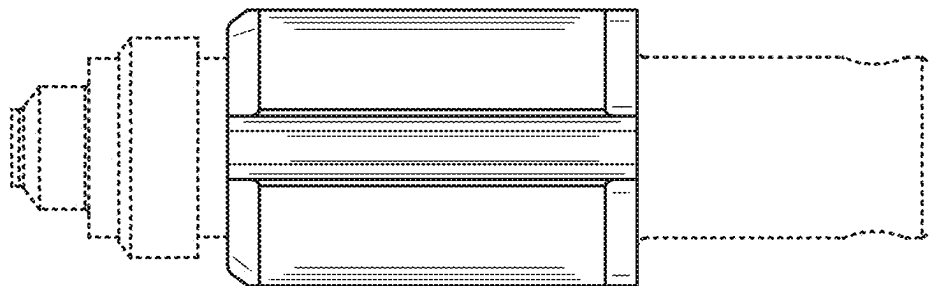


Fig. 5

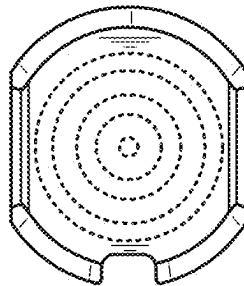


Fig. 6

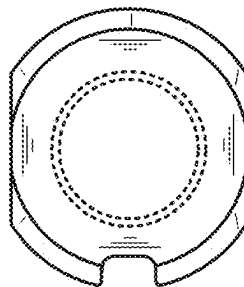


Fig. 7